Computer Aided Geometric Design 8 (1991) 217–221
North-Holland

Local surface interpolation with Bézier patches: errata and improvements

Leon A. Shirman and Carlo H. Séquin

Computer Science Division, Department of Electrical Engineering and Computer Sciences, University of California, Berkeley, CA 94720, USA

Received September 1990

Abstract

Shirman, L.A. and C.H. Séquin, Local surface interpolation with Bézier patches: errata and improvements, Computer Aided Geometric Design 8 (1991) 217–221.

This brief report corrects errors in the triangular patch subdivision scheme described in [Shirman & Séquin '87]. Furthermore, an improvement of the quadrilateral subdivision scheme discussed in the same paper is presented.

Keywords. Surface modeling, Bézier patches, geometric continuity, local interpolation.

1. Correction of triangular subdivision scheme

In the original paper [Shirman & Séquin '87], a process for replacing a quartic triangular Gregory patch with three G^1 continuous Bézier subpatches has been presented. The conditions for geometric continuity between quartic triangular patches were taken from [Farin '82]. These conditions imply that on the patch boundaries, *quartic* control points of degree-elevated cubic boundaries are to be used. However, *cubic* control points were erroneously used in our paper. As a result, the constructed patches would not be G^1 continuous across the boundaries.

Thus, Fig. 3.1 of the original paper should be replaced by Fig. 1 below, so that vectors T_0 and T_3 point to quartic control points of degree-elevated boundaries. Furthermore, Fig. 4.1 of the original paper should be replaced with Fig. 2. In this figure, points with superscripts q correspond to quartic control points on internal and external boundaries.

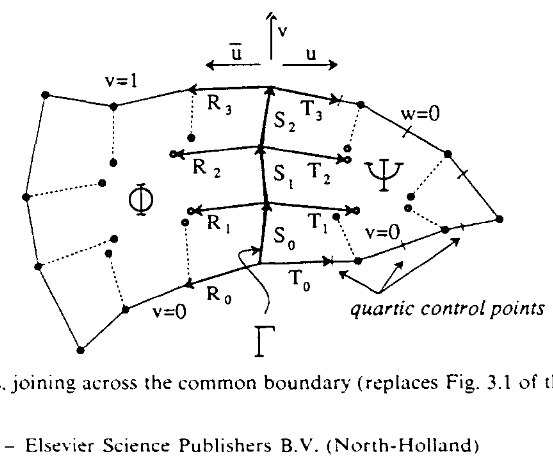

Fig. 1. Two patches, joining across the common boundary (replaces Fig. 3.1 of the original paper).

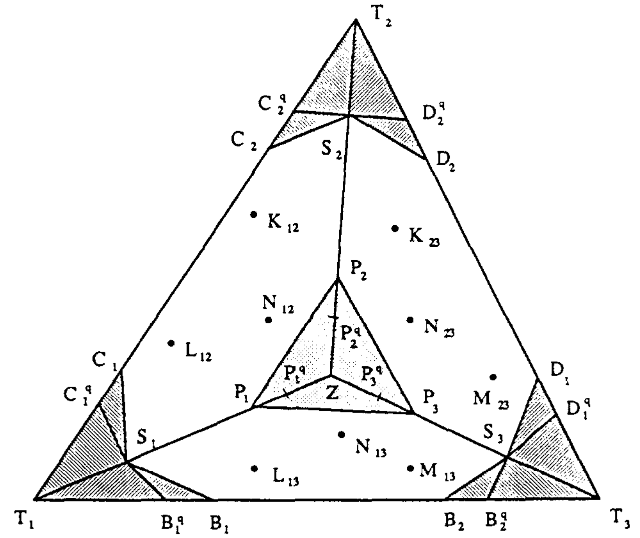

Fig. 2. Triangular patch subdivision (replaces Fig. 4.1 of the original paper).

Omitting intermediate computations, the final formulas for P_i and N_{ij} at the bottom of p. 286 are to be replaced by:

$$P_1 = \tfrac{1}{6}(T_1 - 3S_1 + 4L_{13} + 4L_{12}),$$

$$P_2 = \tfrac{1}{6}(T_2 - 3S_2 + 4K_{23} + 4K_{12}),$$

$$P_3 = \tfrac{1}{6}(T_3 - 3S_3 + 4M_{23} + 4M_{13}),$$

$$N_{12} = \tfrac{1}{4}(-S_1 - S_2 + S_3 + 4P_1 + 4P_2 - 3P_3),$$

$$N_{13} = \tfrac{1}{4}(-S_1 + S_2 - S_3 + 4P_1 - 3P_2 + 4P_3),$$

$$N_{23} = \tfrac{1}{4}(S_1 - S_2 - S_3 - 3P_1 + 4P_2 + 4P_3).$$

The subdivision of a regular n-gon is also affected. The final formula for $N_{i,i+1}$ at the end of Section 4.3 should be modified as follows:

$$N_{i,i+1} = \frac{1}{4(\beta+2)}\big((6-\beta)(P_i + P_{i+1}) + 2\beta(S_i + S_{i+1})$$
$$+(-\beta^2 + 4\beta - 4)Z + (\beta^2 - 2\beta)S\big).$$

Finally, since quadrilateral subdivision scheme 'a' (Section 5) uses triangular patches, the formulas at the top of p. 292 should also be changed:

$$P_1 = \tfrac{1}{4}(4L_{12} + 2M_{14} - 3S_1 + T_1),$$

$$N_{12} = \tfrac{1}{144}(-36S_1 - 36S_2 + 16Q_2 + 139P_1 + 139P_2 - 78Q_1),$$

$$N_{14} = \tfrac{1}{72}(-36S_1 + 36S_2 - 16Q_2 + 101P_1 - 107P_2 + 94Q_1),$$

$$N_{23} = \tfrac{1}{72}(36S_1 - 36S_2 - 16Q_2 - 107P_1 + 101P_2 + 94Q_1).$$

The formulas for subdivision scheme 'b' do not require revision, because this scheme uses only quadrilateral patches.

2. Improvement of quadrilateral subdivision

Although the quadrilateral subdivision scheme 'b' (Section 5 of the original paper) did produce G^1 patches, T. Takamura at Ricoh Corp. pointed out that under certain circumstances the resulting surface was of poor visual quality in the sense that it exhibited unnecessary waves or ripples. The modified approach discussed below drastically improves the appearance of the surface and at the same time maintains overall G^1 continuity. The main idea is to construct Bézier patches so that they closely approximate the original Gregory patch. The justification for this is the fact that interpolation with Gregory patches generally gives good results; one would like to combine the smoothness it produces with the piecewise polynomial surface given by Bézier patches.

We have tested two general approaches to the subdivision process on the scheme 'b'. In the first approach, several of the available degrees of freedom were used to ensure that the 4 *interior network joints* Z_i (Fig. 3) lie on the surface of the Gregory patch. In the second approach, the *normals* at the points Z_i's were required to coincide with corresponding normals of the Gregory patch. A number of test cases clearly indicated that the interpolation of the normals consistently produced more pleasing, smooth surfaces compared to point interpolation.

We now describe in detail the process of determining the coordinates of the 4 interior joints Z_i and the 8 control points Q_{ij} and V_{ij} so that the normals at Z_i coincide with the normals n_i of the Gregory patch at parametric values $(\frac{1}{3}, \frac{1}{3})$, $(\frac{1}{3}, \frac{2}{3})$, $(\frac{2}{3}, \frac{2}{3})$, and $(\frac{2}{3}, \frac{1}{3})$ respectively. The computation proceeds as follows:

Step 1. Compute the interior control points of the Gregory patch according to [Chiyokura '86]. Evaluate the resulting Gregory patch at the 4 center points (with above parametric values) yielding 4 points Z_i^0, and compute the surface normals there yielding the 4 normals n_i.

Step 2. Compute the points P_i as before:

$$P_1 = \tfrac{1}{6}(3L_{12} + 3M_{14} - S_1 + T_1).$$

Step 3. Select the joints Z_i on a line through Z_i^0 with direction n_i, so that the line segment P_iZ_i lies in a plane with the normal n_i, i.e. in the tangent plane of the surface at Z_i (Fig. 4):

$$Z_i = Z_i^0 - \left(\left(P_i - Z_i^0\right) \cdot n_i\right) n_i.$$

Step 4. At this stage, the 4 center points Z_i can be viewed as additional vertices with prescribed normals n_i that need to be interpolated. Thus, any reasonable procedure for finding the control points of a cubic boundary can be used. For example, Q_{14}^0 could be determined by

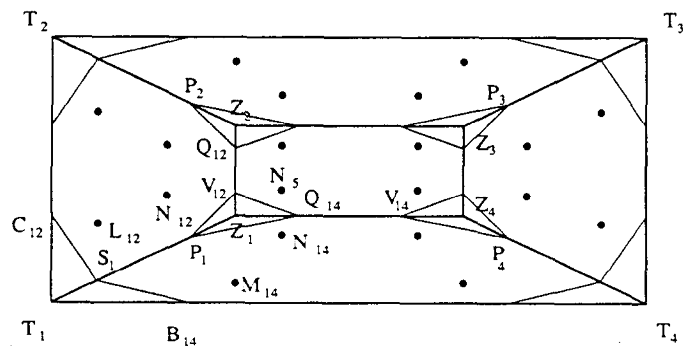

Fig. 3. Quadrilateral patch subdivision for the scheme 'b' (same as Fig. 5.1b of the original paper).

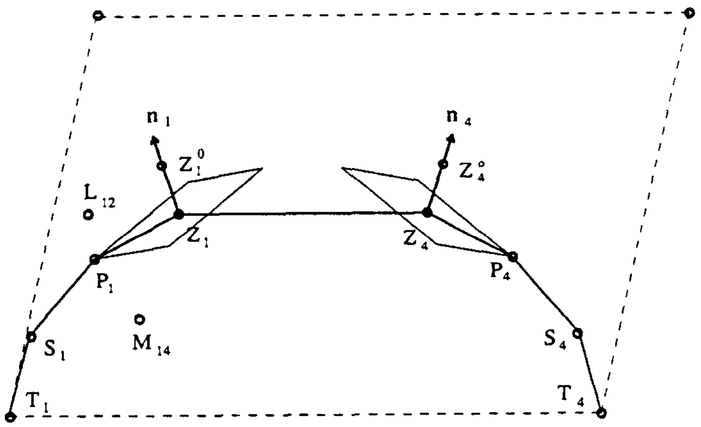

Fig. 4. Computation of interior control points of quadrilateral subdivision.

projecting the edge Z_1Z_4 onto the tangent plane at Z_1, defined by $\boldsymbol{n}_1$, and taking $\frac{1}{3}$ of the length of Z_1Z_4 in the direction of this projection:

$$Q_{14}^0 = Z_1 + \tfrac{1}{3}\,|Z_4 - Z_1|\,\frac{\mathrm{proj}_{\boldsymbol{n}_1}(Z_4 - Z_1)}{|\mathrm{proj}_{\boldsymbol{n}_1}(Z_4 - Z_1)|},$$

$$V_{12}^0 = Z_1 + \tfrac{1}{3}\,|Z_2 - Z_1|\,\frac{\mathrm{proj}_{\boldsymbol{n}_1}(Z_2 - Z_1)}{|\mathrm{proj}_{\boldsymbol{n}_1}(Z_2 - Z_1)|}.$$

Step 5. To guarantee G^1 continuity across interior boundaries, control points are adjusted as follows:

$$Q_{14} = \tfrac{1}{2}\left(3Z_1 - P_1 + Q_{14}^0 - V_{12}^0\right),$$
$$V_{12} = \tfrac{1}{2}\left(3Z_1 - P_1 - Q_{14}^0 + V_{12}^0\right).$$

This would place Z_1 in the center of gravity of the triangle $P_1Q_{14}V_{12}$. Thus, the area ratio constraint [Farin '82], which is a necessary condition for G^1 continuity between the subpatches, will be automatically satisfied across all 8 interior boundaries.

Step 6. Finally, the remaining interior control points of the Bézier patches are computed according to the formulas in the original paper (p. 292):

$$N_{12} = \tfrac{1}{3}(-S_1 - Q_{12} + V_{14} + 3P_1 + 5V_{12} - 4Q_{14}),$$
$$N_{14} = \tfrac{1}{3}(-S_1 + Q_{12} - V_{14} + 3P_1 - 4V_{12} + 5Q_{14}),$$
$$N_5 = \tfrac{1}{9}(3S_1 - 3Q_{12} - 3V_{14} - 10P_1 + 11V_{12} + 11Q_{14}).$$

The above method has been thoroughly tested on a number of well-behaved curve meshes as well as on meshes with highly warped boundaries. In all cases examined, the described approach produced superior results compared to the previous one [Shirman & Séquin '87] in terms of visual quality of the surface. At present, the new method is also implemented with very good results in the DESIGNBASE modeling system of Ricoh Corp [Chiyokura '88].

References

Barnhill, R. and Riesenfeld, W. eds. (1974). *Computer Aided Geometric Design*, Academic Press, New York.

Chiyokura, H. (1986). Localized surface interpolation method for irregular meshes, *Advanced Computer Graphics*, Springer, Tokyo, 3–19.

Chiyokura, H. (1988), *Solid Modeling with DESIGNBASE*, Addison-Wesley, Reading, MA.

Farin, G. (1982), A construction for visual continuity of polynomial surface patches, Computer Graphics and Image Processing 20, 272–282.

Gregory, J. (1974), Smooth interpolation without twist constraints, in: [Barnhill & Riesenfeld '74].

Shirman, L. and Séquin, C. (1987), Local surface interpolation with Bézier patches, Computer Aided Geometric Design 4, 279–295.